NUMERICAL COMPUTING II
ASSIGNMENT VII
Composite Trapezoidal Rule

Solutions, calculations, numerical results, plots, and conclusions

Formula used

For n equal subintervals on [a,b], the step size is h=(b−a)/n. The composite trapezoidal approximation is

$$T_n = h\,[\, f(x_0)/2 + f(x_1) + \cdots + f(x_{n-1}) + f(x_n)/2 \,]$$

The endpoints have weight 1/2; all interior data values have weight 1. The absolute error is $|I_{exact} - T_n|$ when the exact integral is known, or $|I_{reference} - T_n|$ when a highly accurate reference value is used.

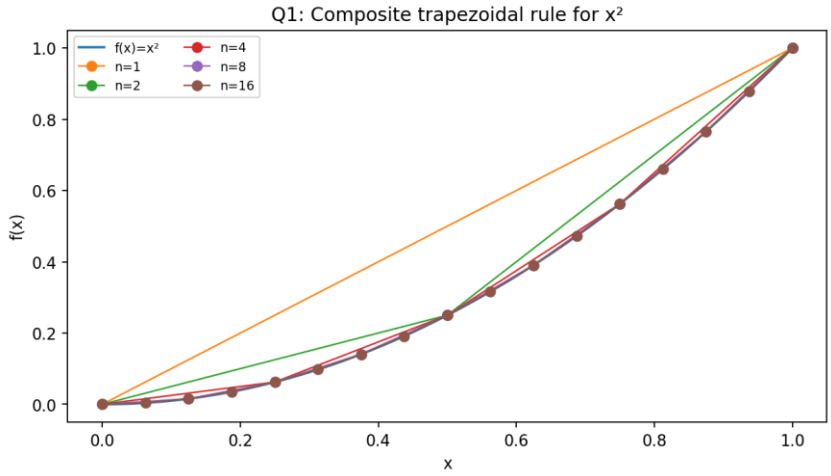
Question 1. Composite trapezoidal rule for x^2

Evaluate $I = \int_0^1 x^2\,dx$ using n = 1, 2, 4, 8, 16.

The exact value is $I = [x^3/3]_0^1 = 1/3 \approx 0.333333333333$.

For example, with n=2: h=(1−0)/2=0.5 and $T_2 = 0.5[(0^2)/2 + 0.5^2 + (1^2)/2] = 0.375$.

n	h	Trapezoidal approximation T_n	Absolute error $\lvert 1/3 - T_n \rvert$
1	1	0.5	0.166666666667
2	0.5	0.375	0.0416666666667
4	0.25	0.34375	0.0104166666667
8	0.125	0.3359375	0.00260416666667
16	0.0625	0.333984375	0.000651041666667



Observation: as n increases, h decreases and the approximation approaches the exact value 1/3 from above. For this convex function, the straight-line trapezoids lie above the curve, so the trapezoidal

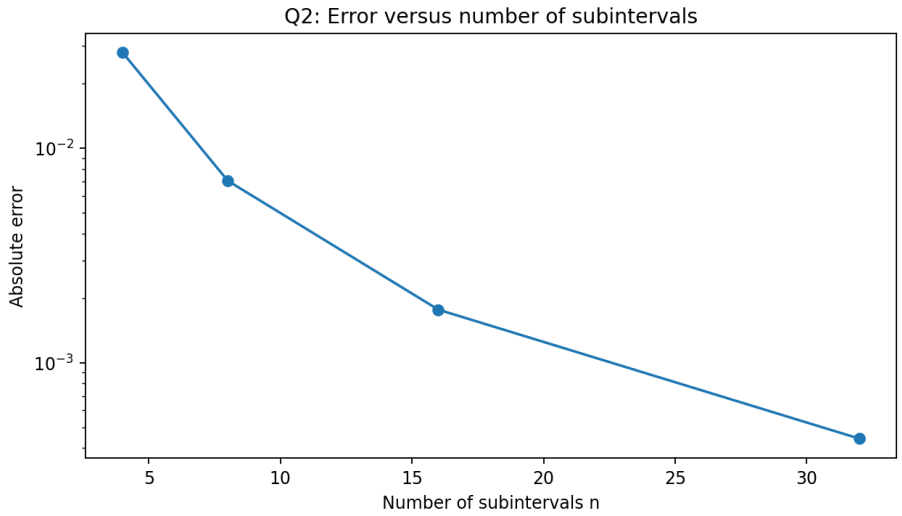
approximation overestimates the integral. The error decreases by a factor of about four whenever n doubles, consistent with the second-order accuracy of the composite trapezoidal rule for a sufficiently smooth function.

Question 2. A function without an elementary antiderivative

Evaluate $I = \int_0^1 e^{(x^2)}\,dx$ using n = 4, 8, 16, 32. The assignment asks for a sufficiently accurate reference value rather than an elementary antiderivative.

For comparison, a high-resolution composite Simpson calculation gives the reference value $I \approx 1.462651745907$. This is a numerical reference value, not an elementary closed-form antiderivative.

n	**h**	**Trapezoidal approximation T_n**	**Absolute error $\lvert Iref - T_n \rvert$**	**$T_n - Iref$**
4	0.25	1.4906788617	0.0280271157917	0.0280271157917
8	0.125	1.46971227643	0.00706053052248	0.00706053052248
16	0.0625	1.46442031015	0.0017685642423	0.0017685642423
32	0.03125	1.46309410261	0.000442356699246	0.000442356699246



Observation: for these n values, the approximations increase toward the reference value from below, and the absolute error decreases as n increases. Increasing n generally improves this calculation because the subintervals become narrower and the curved function is represented more accurately by local straight-line segments. However, it is not a universal guarantee for every possible function, node arrangement, or floating-point computation.

Question 3. Data instead of a function

Given experimental data:

x	**0**	**0.5**	**1.0**	**1.5**	**2.0**
f(x)	1.00	1.65	2.70	4.48	7.39

Estimate $\int_0^2 f(x)\,dx$ using the composite trapezoidal rule. The spacing is uniform: h=0.5.

$T_4 = h[f(0)/2 + f(0.5) + f(1.0) + f(1.5) + f(2.0)/2]$

$T_4 = 0.5[1.00/2 + 1.65 + 2.70 + 4.48 + 7.39/2] = 6.512500.$

Therefore, the estimated integral is approximately 6.512500.

Why is numerical integration useful here? The available information consists of measured or tabulated values rather than a known formula for f(x). An antiderivative cannot be derived from the table alone, but the trapezoidal rule can estimate the accumulated area using the data points. The estimate's accuracy depends on the spacing of measurements and how well straight-line segments represent the unknown function between samples.

Overall conclusion

The composite trapezoidal rule estimates definite integrals using function values at discrete points. Increasing the number of subintervals usually improves the approximation for smooth functions by reducing the step size. For x^2, the exact integral allows direct error calculation. For e^(x^2), a high-accuracy numerical reference is used. For experimental data, the rule remains applicable even when no explicit function or antiderivative is known.

Code Python program

```
import math

def composite_trapezoidal(f, a, b, n):
    if n <= 0:
        raise ValueError("n must be a positive integer")
    h = (b - a) / n
    total = 0.5 * f(a) + 0.5 * f(b)
    for i in range(1, n):
        total += f(a + i * h)
    return h, h * total

# Question 1: integral of x^2 from 0 to 1
f1 = lambda x: x**2
exact1 = 1.0 / 3.0
print("Q1: integral of x^2")
for n in [1, 2, 4, 8, 16]:
    h, approx = composite_trapezoidal(f1, 0.0, 1.0, n)
    print(f"n={n:2d}, h={h:.6f}, T_n={approx:.12f}, "
          f"absolute error={abs(exact1-approx):.12g}")

# Question 2: integral of exp(x^2) from 0 to 1
f2 = lambda x: math.exp(x*x)
# High-accuracy reference calculated separately using a fine Simpson rule
reference = 1.462651745907  # rounded numerical reference
print("\\nQ2: integral of exp(x^2)")
for n in [4, 8, 16, 32]:
    h, approx = composite_trapezoidal(f2, 0.0, 1.0, n)
    print(f"n={n:2d}, h={h:.6f}, T_n={approx:.12f}, "
          f"absolute error={abs(reference-approx):.12g}")

# Question 3: experimental data
x = [0.0, 0.5, 1.0, 1.5, 2.0]
y = [1.00, 1.65, 2.70, 4.48, 7.39]
h = x[1] - x[0]
approx = h * (0.5*y[0] + sum(y[1:-1]) + 0.5*y[-1])
print("\\nQ3: experimental data")
```

```
print(f"h={h:.3f}, trapezoidal estimate={approx:.6f}")
```

```
======================================================================
PROBLEM 1: Integral of x^2 from 0 to 1
======================================================================

Required calculations:
----------------------------------------------------------------------
Number of intervals     Approximation           Absolute Error          Accuracy
----------------------------------------------------------------------
1                       0.50000000              0.16666667              50.000000%
2                       0.37500000              0.04166667              87.500000%
4                       0.34375000              0.01041667              96.875000%
8                       0.33593750              0.00260417              99.218750%
16                      0.33398438              0.00065104              99.804688%

Exact value = 0.3333333333333333

Enter another number of intervals for Problem 1 (0 to continue): 8

For 8 intervals:
Approximation = 0.3359375
Absolute error = 0.002604166666666685
Accuracy = 99.21875 %

Enter another number of intervals for Problem 1 (0 to continue): 0
```

```
======================================================================
PROBLEM 2: Integral of exp(-x^2) from 0 to 1
======================================================================

Required calculations:
----------------------------------------------------------------------
Number of intervals     Approximation           Absolute Error          Accuracy
----------------------------------------------------------------------
4                       0.74298410              0.00384004              99.485818%
8                       0.74586561              0.00095852              99.871654%
16                      0.74658460              0.00023954              99.967926%
32                      0.74676425              0.00005988              99.991982%

Reference value = 0.746824132806287

Enter another number of intervals for Problem 2 (0 to continue): 0
```

```
======================================================================
PROBLEM 3: Experimental Data
======================================================================

Experimental data:
------------------------------
x = 0       f(x) = 1.0
x = 0.5     f(x) = 1.65
x = 1.0     f(x) = 2.7
x = 1.5     f(x) = 4.48
x = 2.0     f(x) = 7.39

Width between data points = 0.5
Estimated integral = 6.5125
```